

PORT ARTHUR PALMER, Australia

WMO: 949780

Lat: 43.167S

Lon: 147.833E

Elev: 193

StdP: 99.03

Time Zone: 10.00 (AUS)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	4.1	-2.0	3.3	6.4	-1.0	3.5	6.2	13.4	9.3	13.0	8.3	5.5	250	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	24.7	16.4	22.4	15.8	20.4	15.2	17.9	22.6	16.9	20.8	15.9	19.4	5.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.8	19.5	15.3	11.1	18.8	14.2	10.4	17.8	51.3	22.7	48.1	20.8	45.3	19.5	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.1	11.6	10.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	0.5	20.4	0.6	1.0	0.1	21.2	-0.3	21.8	-0.6	22.3	-1.0	23.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	14.3	15.0	13.8	12.1	10.5	8.6	8.4	8.4	9.5	10.5	11.2	12.7		
	DBStd	3.31	2.89	3.09	2.51	2.46	2.07	1.96	1.76	2.13	2.35	2.69	2.52	2.69		
	HDD10.0	265	1	0	1	6	18	50	55	59	36	23	14	3		
	HDD18.3	2618	131	103	143	187	243	291	308	308	266	243	216	179		
	CDD10.0	713	134	139	118	69	33	9	6	9	21	39	48	86		
	CDD18.3	23	6	9	3	1	0	0	0	0	0	1	1	3		
	CDH23.3	134	41	41	16	0	0	0	0	0	0	0	2	34		
	CDH26.7	36	9	12	4	0	0	0	0	0	0	0	0	11		
(14) Wind	WSAvg	5.2	5.2	5.1	5.0	5.2	4.8	4.9	5.1	5.6	5.3	5.2	5.7	5.3		
(15) Precipitation	PrecAvg	1165	89	64	92	93	85	110	111	125	100	103	83	99		
	PrecMax	1535	201	175	313	238	154	226	179	253	202	188	172	328		
	PrecMin	943	19	15	30	41	14	42	47	29	30	39	18	16		
	PrecStd	151	49	45	56	51	39	50	36	55	45	37	38	64		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.3	28.3	26.2	21.9	18.7	14.9	14.1	16.1	18.3	21.7	24.0	27.9		
		MCWB	18.1	17.4	16.8	14.6	12.3	11.3	9.9	10.1	12.8	13.9	15.1	16.6		
	2%	DB	24.0	24.0	21.4	19.5	16.5	13.6	12.8	14.2	16.1	19.3	19.8	22.6		
		MCWB	17.0	16.8	15.1	13.9	12.3	10.7	9.5	9.6	11.7	14.1	13.2	15.6		
	5%	DB	21.1	21.9	19.4	17.7	15.2	12.7	12.0	12.9	14.4	16.8	17.4	19.2		
		MCWB	16.1	16.5	14.6	13.2	11.6	10.0	9.2	8.9	10.0	12.1	12.4	14.6		
	10%	DB	18.9	19.6	17.8	16.2	14.0	11.8	11.3	12.0	13.1	14.7	15.4	17.1		
		MCWB	14.9	15.6	14.2	12.5	10.5	8.9	8.6	8.5	9.4	10.8	11.6	13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.6	17.9	16.0	14.2	12.2	11.6	11.9	13.8	16.7	16.0	17.7		
		MCDB	26.1	24.3	22.0	19.5	16.5	13.9	12.2	13.5	16.5	19.1	22.0	23.5		
	2%	WB	17.7	18.0	16.4	14.9	13.1	11.3	10.5	10.4	12.1	13.8	14.2	16.4		
		MCDB	22.5	22.6	19.2	18.2	15.8	13.2	12.0	12.8	15.3	18.3	18.4	21.3		
	5%	WB	16.5	17.0	15.5	13.9	11.8	10.2	9.7	9.5	10.7	12.7	13.2	15.0		
		MCDB	20.5	20.6	18.5	16.9	14.9	12.4	11.6	12.3	13.6	16.1	16.6	18.7		
	10%	WB	15.2	16.0	14.5	12.8	11.0	9.2	8.9	8.7	9.7	11.5	12.1	13.7		
		MCDB	18.5	19.1	17.8	15.8	13.7	11.3	11.0	11.5	12.6	14.5	14.9	17.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.3	5.7	5.3	4.3	3.7	3.7	4.7	5.2	5.7	5.7	5.7		
		MCDBR	10.1	9.7	8.5	7.3	5.9	4.5	4.4	6.2	7.0	8.8	9.1	9.6		
		MCWBR	5.7	5.5	4.8	4.5	4.0	3.6	3.4	4.0	4.6	5.6	5.9	5.9		
	5% WB	MCDBR	9.1	8.7	7.3	6.6	5.5	4.1	4.0	5.6	6.3	7.7	8.0	8.5		
		MCWBR	5.7	5.7	4.8	4.6	4.1	3.6	3.4	4.0	4.7	5.4	5.6	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.339	0.330	0.325	0.309	0.297	0.291	0.294	0.304	0.325	0.326	0.328	0.333		
	taud		2.552	2.594	2.604	2.622	2.629	2.618	2.616	2.599	2.531	2.548	2.563	2.560		
	Ebn at Noon		986	968	927	872	813	779	806	864	910	962	991	998		
	Edh at Noon		105	96	86	72	60	55	59	72	91	99	103	106		
(44) All-Sky Solar Radiation	RadAvg		6.41	5.44	4.05	2.76	1.70	1.35	1.57	2.35	3.56	4.97	5.83	6.55		
	RadStd		0.44	0.29	0.21	0.16	0.11	0.10	0.07	0.15	0.17	0.25	0.44	0.34		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+0.34	N/A	N/A	+0.71	N/A	N/A	N/A	-106	+110	+17

Nomenclature: See separate page